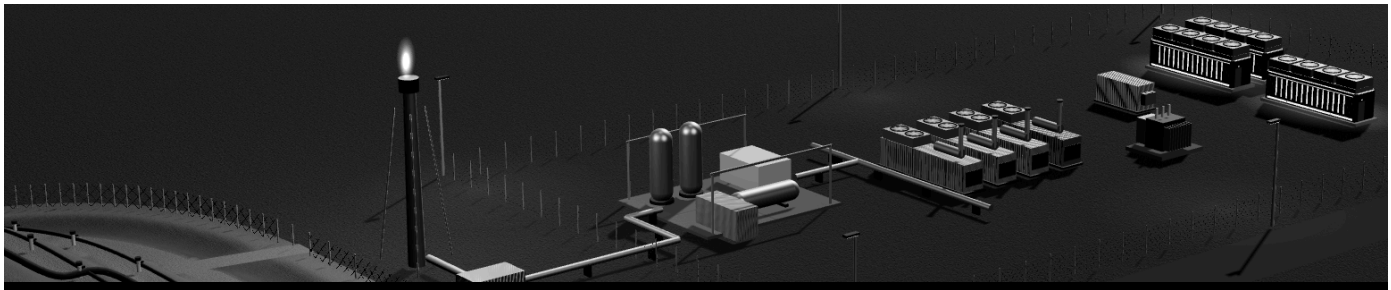
Mostoller Landfill, Somerset, Pennsylvania: a first sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

3.4 MW	3	9 ac	312k
POWER FROM THE GAS	COMPUTE MODULES	LAND NEEDED	MMBTU BOUGHT / YR

GAS SALE, PER YEAR	LAND LEASE, PER YEAR	TOTAL TO THE OWNER, PER YEAR
up to $234k	$9k to $18k	$9k to $252k



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=1186&CH4=52.4

THE MATH

Landfill gas collected and flared (2023)	1,186 scfm
Methane content (2023)	52.4%
Energy in the gas	37.4 MMBtu per hour
Power at a heat rate of 11,000 Btu/kWh	about 3.4 MW
Compute (power less cooling and losses, /1.2)	about 2.8 MW
Compute modules (about 1 MW each)	3
Gas engines (running plus one spare)	4 at 1.6 MW
Land: 5 acres plus 1.2 acres per MW	about 9 acres
Gas bought per year (95% run time)	312k MMBtu
Gas payment, up to $0.75/MMBtu	up to $234k
Land lease at $1,000 to $2,000 per acre per year	$9k to $18k

Landfill gas: about 1,012 Btu per cubic foot of methane. Prices in US dollars; the price band is the one shown on our Site model page.

WHAT IT WOULD LOOK LIKE

- Gas engines on a small parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site. The owner puts in no capital.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

WORTH CHECKING

- Gas rights and any existing gas agreement
- PA DEP air plan approval (25 Pa. Code Chapter 127)
- Tie-in under the landfill's Title V permit (NSPS Subpart XXX, NESHAP Subpart AAAA)
- County or municipal zoning for the data hall

Source: EPA Greenhouse Gas Reporting Program, 2023: about 1,186 scfm collected at 52.4% methane, with no energy project on record. This is a first read from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.